

Chapter no 5

5.1 Heat Engines and Refrigerators

Heat Engines: A heat engine is a device that converts heat energy into mechanical work.

According to 2nd law of thermodynamics

$$W = Q_H - Q_L$$

$$\eta (\text{thermal efficiency}) = \frac{W}{Q_H} = \frac{Q_H - Q_L}{Q_H} = 1 - \frac{Q_L}{Q_H}$$

For an ideal Carnot cycle

$$\eta = 1 - \frac{T_L}{T_H}$$

Refrigerators: A refrigerator is a device that transfers heat from a low temp reservoir to a high temp reservoir.

According to 2nd law of thermodynamics

$$Q_H = W + Q_L$$

→ For refrigerator:

$$\text{COP}_{\text{ref}} = \frac{Q_L}{W}$$

↳ For ideal Carnot cycles

$$\text{COP} = \frac{T_L}{T_H - T_L}$$

→ For heat pump:

$$\text{COP} = \frac{Q_H}{W}$$

↳ For ideal Carnot cycle:

$$\text{COP} = \frac{T_H}{T_H - T_L}$$

Example 5.1

$$W = 136 \text{ hp}$$

$$= 136 \times 0.7355 = 100 \text{ kW}$$

(table A.1)

$$\eta = 30\% ; q_H = 35000 \text{ kJ/kg}$$

As we know

$$\eta = \frac{\dot{W}}{\dot{Q}_H} \Rightarrow \dot{Q}_H = \frac{\dot{W}}{\eta}$$

$$\dot{Q}_H = \frac{100 \text{ kW}}{30\%} = \boxed{333 \text{ kW}}$$

$$\text{Now, } \dot{Q}_L = \dot{Q}_H - \dot{W}$$

$$\dot{Q}_L = 333 - 100 = \boxed{233 \text{ kW}}$$

For rate of fuel combustion

$$\dot{m} = \frac{\dot{Q}_H}{q_H} = \frac{333}{35000}$$

$$\dot{m} = \boxed{0.00951 \text{ kg/s}}$$

Example 5.2

$$\dot{W} = 150 \text{ W}, \dot{Q}_H = 400 \text{ W}$$

Energy equation:

$$\dot{Q}_H = \dot{W} + \dot{Q}_L$$

$$\dot{Q}_L = \dot{Q}_H - \dot{W}$$

$$\dot{Q}_L = 400 - 150 = \boxed{250 \text{ W}}$$

$$\text{COP} = \beta = \frac{\dot{Q}_L}{\dot{W}} = \frac{250}{150}$$

$$\boxed{\beta = 1.67}$$

5.2 Second Law of Thermodynamics

2nd law of thermodynamics can be expressed by:

→ Kelvin-Planck Statement

It is impossible to construct a heat engine that operates in a cycle and convert all the heat into work. Some heat must always be rejected to a low temperature reservoir.

→ Clausius Statement

It is impossible to construct a device that operates in a cycle and transfer heat from low to high without any external work being done.

∴ Carnot cycle satisfies both statements by operating under reversible process and achieve the maximum possible efficiency.

5.5 Carnot Cycle

Carnot cycle based on 4 major steps

1- Isothermal Expansion (heat absorption)

The working fluid absorbs heat (Q_H) from high res (T_H). Temp remains constant.

2- Adiabatic Expansion

The gas expands. No heat transfer.
Temp decr from T_H to T_L .

3- Isothermal Compression (Heat rejection)

Working fluid releases heat (Q_L) to a low temp reservoir (T_L). Temp remains constant.

4- Adiabatic Compression

Gas is compressed without heat transfer.
Temp incr from T_L to T_H .

Problem 5.44

$$T_H = 300^\circ\text{C} = 573\text{K} \quad ; \quad T_L = 45^\circ\text{C} = 318\text{K}$$

$$\eta_{\text{Carnot}} = 1 - \frac{T_L}{T_H}$$

$$\text{So, } \eta = 1 - \frac{318}{573}$$

$$\boxed{\eta = 0.44 \text{ or } 44.50\%}$$



Example 5.4

$$\dot{Q}_H = 1000 \text{ kW} = 1 \text{ MW}$$

$$\dot{W} = 450 \text{ kW}; T_L = 300 \text{ K}$$

$$T_H = 550^\circ\text{C} = 823.15 \text{ K}$$

For heat engine:

$$\dot{W} = \dot{Q}_H - \dot{Q}_L$$

$$\dot{Q}_L = \dot{Q}_H - \dot{W}$$

$$\dot{Q}_L = 1000 - 450 = \boxed{550 \text{ kW}}$$

$$\eta = \frac{\dot{W}}{\dot{Q}_H} = \frac{450}{1000} = \boxed{0.45}$$

$$\text{or } \boxed{45\%}$$

For Carnot efficiency

$$\eta = 1 - \frac{T_L}{T_H}$$

$$\eta = 1 - \frac{300}{823.15} = \boxed{0.635}$$

$$\text{or } \boxed{63.55\%}$$

Now the rate of work and heat rejection

$$\dot{W} = (\eta_c)(\dot{Q}_H) = 0.635 \times 1000$$

$$\boxed{\dot{W} = 635 \text{ kW}}$$

and

$$\dot{Q}_L = \dot{Q}_H - \dot{W}$$

$$= 1000 - 635 = \boxed{365 \text{ kW}}$$



Problem 5.65

$$T_1 = 20^\circ\text{C} \text{ and } P_1 = 200 \text{ kPa}$$

the state is superheated vapor.

At these conditions, from table B.5.2

$$v = 0.11436 \text{ m}^3/\text{kg}, u_1 = 395.27 \text{ kJ/kg}$$

$$\text{Now } V = 0.1 \text{ m}^3$$

$$m = \frac{V}{v} = \frac{0.1}{0.11436} = 0.8744 \text{ kg}$$

For final state

$$T_2 = -10^\circ\text{C} \text{ and } P = ?$$

$$P_{\text{sat}} = 201.7 \text{ kPa}$$

$$v_f = 0.000755, v_g = 0.09921$$

$$v_2 = \frac{V}{m} = \frac{0.1}{0.8744} = 0.11435$$

As $v_2 > v_g$ (superheated state)

From superheated table

$$\text{at } T = -10, u_2 = 372.15 \text{ kJ/kg}$$

Now

$$\Delta u = m(u_2 - u_1)$$

$$= (0.8744)(372.15 - 395.27)$$

$$\boxed{\Delta u = -20.21}$$

Now,

$$\text{COP} = \frac{\dot{Q}_L}{\dot{W}}$$

$$\dot{W} = \frac{\dot{Q}_L}{\text{COP}}$$

$$COP_{ref} = \frac{T_L}{T_H - T_L} \quad \text{put values in kelvin}$$

$$COP = \frac{263}{293 - 263} = \boxed{8.77}$$

So,

$$W = \frac{20.21}{8.77} = \boxed{2.304 \text{ kJ/kg}}$$

